

Section 12.6: Cylinders and Quadric Surfaces

Vectors and the Geometry of Space

MTH 310
Calculus III

Conic Sections Are Everywhere

A **conic section** is the curve you get by slicing a double cone with a plane.

Depending on the angle of the cut, you get one of four shapes:

- **Circle** — cut straight across (e.g. the rim of a coffee cup)
- **Ellipse** — cut at a slight tilt (e.g. planetary orbits)
- **Parabola** — cut parallel to the side of the cone (e.g. satellite dishes, headlight reflectors)
- **Hyperbola** — cut steep enough to hit both halves (e.g. cooling towers, sonic booms)

Classifying Conics by Their Equation

The **general second-degree equation** in two variables is:

$$ax^2 + by^2 + cx + dy + e = 0$$

If a and b are not **both** zero, this equation describes a **conic section** (possibly degenerate). We classify by comparing the squared terms:

Squared terms	Shape
Only x^2 or only y^2	Parabola
Both, same sign ($ab > 0$)	Circle ($a = b$) or Ellipse ($a \neq b$)
Both, opposite signs ($ab < 0$)	Hyperbola

Standard Forms of Conics

Definition 1: Standard Forms

- **Circle:** $(x - h)^2 + (y - k)^2 = r^2$
- **Ellipse:** $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$
- **Hyperbola (horizontal):** $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$
- **Hyperbola (vertical):** $-\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$

We get these by **completing the square** on the general equation. The center is at (h, k) .

Example 1: Sketch $y^2 - 9x^2 = 1$

Step 1 — Which way does it open? Compare signs of the squared terms: $+y^2$ and $-x^2$. The **positive** variable (y) wins → opens **up and down**.

Step 2 — Find intercepts. Zero out one variable at a time:

- Set $x = 0$: $y^2 = 1 \rightarrow y = \pm 1$ ✓
- Set $y = 0$: $-9x^2 = 1 \rightarrow$ no real solution ✗

Step 3 — Find asymptotes. For large values, the constant 1 is negligible: $y^2 \approx 9x^2 \rightarrow y \approx \pm 3x$

Example 2: Sketch $x^2 - 4y^2 = 1$

Step 1 — Which way does it open? Compare signs: $+x^2$ and $-y^2$. The **positive** variable (x) wins → opens **left and right**.

Step 2 — Find intercepts. Zero out one variable at a time:

- Set $y = 0$: $x^2 = 1 \rightarrow x = \pm 1$ ✓
- Set $x = 0$: $-4y^2 = 1 \rightarrow$ no real solution ✗

Step 3 — Find asymptotes. For large values, the constant 1 is negligible: $x^2 \approx 4y^2 \rightarrow y \approx \pm \frac{1}{2}x$

Example 3: Identify $x^2 - 4y^2 + 4x + 24y = 33$

Step 1 — Group by variable and factor. $(x^2 + 4x) - 4(y^2 - 6y) = 33$

Step 2 — Complete the square. $(x^2 + 4x + 4) - 4(y^2 - 6y + 9) = 33 + 4 - 36$

Step 3 — Simplify. $(x + 2)^2 - 4(y - 3)^2 = 1$

Step 4 — Standard form. $\frac{(x+2)^2}{1} - \frac{(y-3)^2}{(1/2)^2} = 1$ — **Hyperbola** centered at $(-2, 3)$, opening left and right.

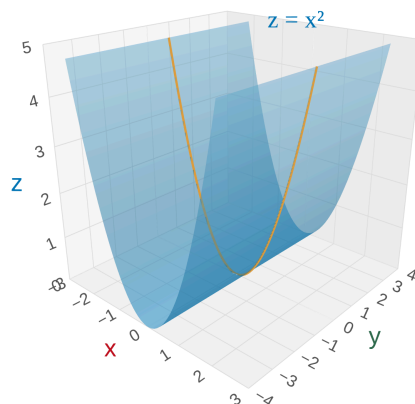
Cylinders in 3D

Definition 2: Cylinder

A **cylinder** is a surface formed by taking a 2D curve and extending it along the missing coordinate axis.

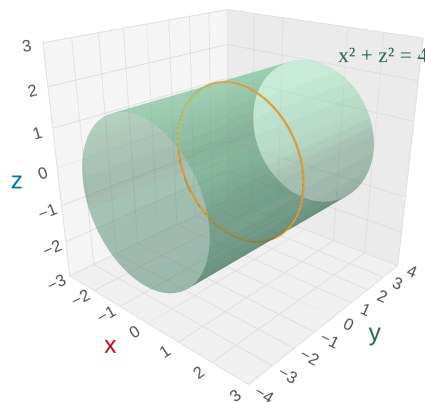
If one variable is absent from the equation, the graph is a cylinder that extends infinitely along that axis.

$z = x^2$ — missing y , so it extends along the y -axis: **parabolic cylinder**



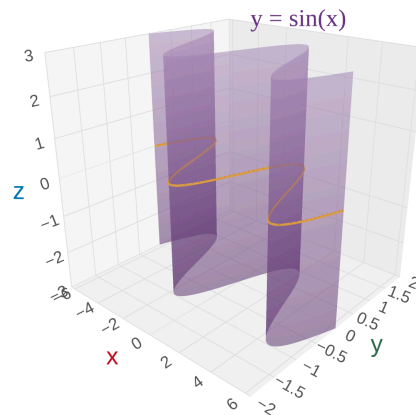
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$x^2 + z^2 = 4$ — missing y , so it extends along the y -axis: **circular cylinder**



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$y = \sin(x)$ — missing z , so it extends along the z -axis: **wave cylinder**



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Key idea: the missing variable is *free* — it can be anything.

Quadric Surfaces and Traces

Definition 3: Quadric Surface

A **quadric surface** is the 3D graph of a second-degree equation in x , y , and z .

Definition 4: Trace

A **trace** is the cross-section you get by setting one variable equal to a constant.

- **Horizontal trace** ($z = k$): slice parallel to the xy -plane
- **Vertical trace** ($x = k$ or $y = k$): slice parallel to a coordinate wall

Strategy: Find traces in all three directions, then assemble the 3D shape.

Example 4: Identify $x^2 + y^2 - z = 0$

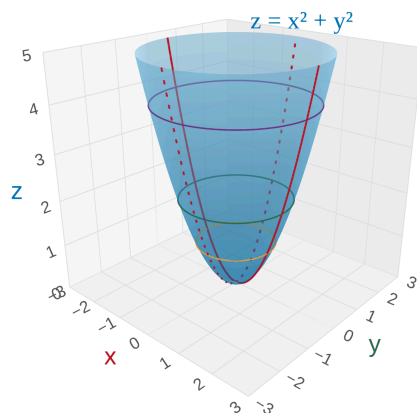
Rewrite as $z = x^2 + y^2$. Now find traces:

Horizontal traces ($z = k$): $x^2 + y^2 = k$

- $k > 0$: **circle** of radius $\sqrt{k}$ — circles get larger as z increases
- $k = 0$: just the origin
- $k < 0$: empty (no surface below $z = 0$)

Vertical traces ($y = k$ or $x = k$): upward **parabolas** shifted up by k^2

We don't need every trace to sketch the surface. Pick the direction with the simplest shape — here the **circles** ($z = k$) — and imagine stacking them. Then one parabola in each vertical direction serves as a guide.



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Elliptic paraboloid (bowl shape) — opens upward along the z -axis

Example 5: Identify $z = y^2 - x^2$

Rewrite as $-x^2 + y^2 - z = 0$. Now find traces:

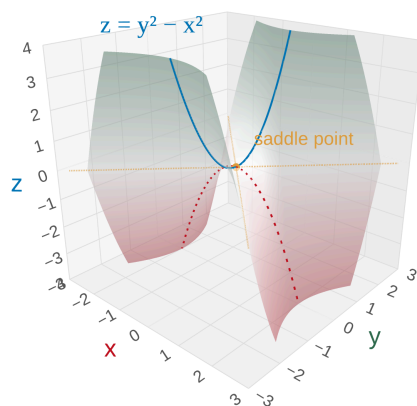
Horizontal traces ($z = k$): $-x^2 + y^2 = k$

- $k > 0$: hyperbola opening along y
- $k = 0$: lines $y = \pm x$ (degenerate)
- $k < 0$: hyperbola opening along x

Vertical traces:

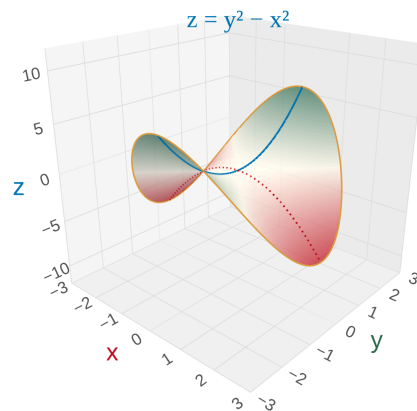
- $x = 0$: $z = y^2$ — upward parabola
- $y = 0$: $z = -x^2$ — downward parabola

Stack the hyperbolas (which rotate 90° as z crosses 0), then use the two parabolas as vertical guides — the surface curves **up** in y and **down** in x .



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Hyperbolic paraboloid (saddle / "Pringle" shape)



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Example 6: Identify $-x^2 + y^2 + z^2 = 1$

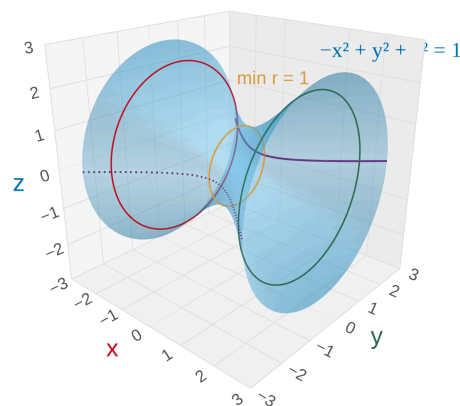
Find traces:

Simplest trace ($x = k$): $y^2 + z^2 = 1 + k^2$ — **circles** of radius $\sqrt{1 + k^2}$

- Always ≥ 1 , so circles exist for **all** k (no gaps!)
- Minimum radius 1 at $x = 0$ (the "waist")

Vertical traces ($z = 0$): $-x^2 + y^2 = 1$ — **hyperbola** opening along y

Stack the circles along the x -axis — they shrink to a minimum at $x = 0$ and expand outward in both directions. The hyperbola in the xy -plane confirms the waist shape.



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Hyperboloid of one sheet — wraps continuously around the x -axis

Example 7: Identify $-x^2 - y^2 + z^2 = 1$

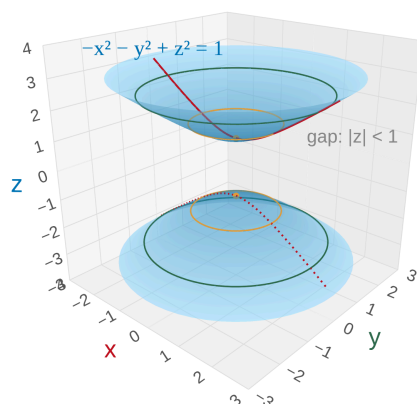
Find traces:

Horizontal traces ($z = k$): $x^2 + y^2 = k^2 - 1$

- $|k| > 1$: **circle** of radius $\sqrt{k^2 - 1}$
- $|k| = 1$: just a point (the vertex of each sheet)
- $|k| < 1$: **empty** — no surface in the gap!

Vertical trace ($y = 0$): $-x^2 + z^2 = 1$ — **hyperbola** opening along z

Stack circles above $z = 1$ and below $z = -1$, expanding outward. The gap $|z| < 1$ means we get **two separate sheets**. The hyperbola in the xz -plane serves as a vertical guide.



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Hyperboloid of two sheets — two separate pieces with a gap between them

Example 8: Identify $z^2 = x^2 + y^2$

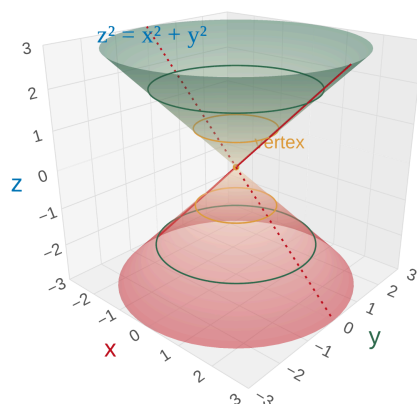
Find traces:

Horizontal traces ($z = k$): $x^2 + y^2 = k^2$

- $k \neq 0$: **circle** of radius $|k|$
- $k = 0$: just the origin (the vertex)

Vertical trace ($y = 0$): $z^2 = x^2 \rightarrow z = \pm x$ — two **lines** through the origin

Stack circles that grow linearly with $|z|$. Unlike the paraboloid (circles grow like $\sqrt{z}$), these grow at a constant rate — straight sides. The two lines confirm it: no curvature, just a cone.



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Cone — the boundary case between hyperboloids of one and two sheets

Homework

Section 12.6: 1, 3, 6, 8, 13, 17, 21, 23–30, 47, 49